

Train, Validation, Test

Project: Responsible Plant Classifier

1. Train, Validation, Test

Teach the learning cycle

Primary target: Select a model using a declared validation rule

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Hold Out Test Data

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Absolute Error and MAE

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate MAE MSE and RMSE

Reference example, not the only acceptable wording: Teacher explanation: $(1+2+3)/3=2$ cm.

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Categorize development errors using evidence

Reference example, not the only acceptable wording: Teacher explanation: Dim 4/12 about 33.3 percent; bright 3/18 about 16.7 percent.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

You have 30 leaves with 3 views each. Allocate 18 leaves to train, 6 to validation, 6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Concept to explain

Training fits parameters, validation chooses settings, and testing estimates final performance after choices are frozen.

Term · Meaning

Term: training set

Meaning: data used to fit parameters

Term: validation set

Meaning: data used to choose settings

Term: test set

Meaning: data used for final estimate

Term: split strategy

Meaning: rule assigning examples to sets

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on validation and model selection. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Training rows fitted parameters; separate validation rows compared performance.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain train, validation, test using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA Training fits parameters, validation chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Fit two candidate trees on training rows. Choose depth and threshold using validation metrics and error slices.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Fit two candidate trees on training rows.

STEP 2 Choose depth and threshold using validation metrics and error slices.

STEP 3 Freeze preprocessing, features, model, and decision threshold.

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Split 100 independent leaves into 60 train, 20 validation, 20 test.

Explain roles and preprocessing.

Exact explanation to model

Train fits parameters and learned transforms; validation chooses settings; final test evaluates the selected system. Group related records first.

Explain the added worked example

Fitting and selecting answer different questions. Training adjusts parameters. Validation helps choose settings or model families. A final test estimates the performance of the frozen development process on held-out examples. Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines validation and model selection.

Three candidates have training MAE A=1, B=0, C=2 and validation MAE A=3, B=5, C=3. The declared rule

is smallest validation MAE, then simpler model, with A simpler than C. Choose A. B fits training perfectly but does worse on validation. The final test remains unopened.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Key terms

Term · Meaning

Term: Validation

Meaning: Development evidence used to choose between options

Term: Hyperparameter

Meaning: A model setting chosen outside its fitted parameters

Term: Selection rule

Meaning: A declared method for choosing an option

Term: Test set

Meaning: Held-out evidence reserved for a final evaluation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Validation and model selection at a glance

Part · Evidence or action · Interpretation

Part: Training

Evidence or action: Fit each candidate

Interpretation: No test labels

Part: Validation

Evidence or action: Choose A by rule

Interpretation: Record every attempt

Part: Final test

Evidence or action: Frozen model only

Interpretation: Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train, Validation, Test

Assign a separate job to training, validation and final test evidence.

The facts we will use

Candidates A and B have training MAE1 and 0 cm; validation MAE2 and 3 cm. Selection rule is smallest validation MAE.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/24

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because 2 cm is below 3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE 2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Fit on training

What happens and why: Both candidates are built using training data; B fits it more closely.

Step: 2. Select on validation

What happens and why: Choose A because 2 cm is below 3 cm.

Step: 3. Reserve final test

What happens and why: Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Repeated validation use can itself overfit model choices; keep a genuinely untouched final evaluation.

Change one thing in the pictured case

Change: A and B both have validation MAE 2 cm; A is simpler under a predeclared tie rule.

Prediction: Select A using the predeclared simpler-model tie-break.

Reason: The validation errors tie, so the stated secondary rule decides without consulting final data.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Split 100 independent leaves into 60 train, 20 validation, 20 test.

Write the question to explain: Explain roles and preprocessing.

Reveal after predictions: Train fits parameters and learned transforms; validation chooses settings; final test evaluates the selected system. Group related records first.

Keep for an independent attempt: Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, validation and test?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Fit on training, select on validation, evaluate the frozen choice on held-out test; fit preprocessing only on training. Exit response: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Which data fitted the line last week? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify validation in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: choose a candidate; assign data roles. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between validation and hyperparameter. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: A team creates D after reading validation mistakes. What changes in the record?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Select a model using a declared validation rule; Keep training, validation and test roles separate. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Where should parameter fitting, setting selection and final reporting occur?

Teacher check: Teacher explanation: Fit on training, select on validation, evaluate the frozen choice on held-out test; fit preprocessing only on training.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Validation and model selection. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why does B's zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Model selection is based on held-out validation error here. Lower training error can reflect fitting the training details rather than better new-case prediction. Changed-case check: Select A using the predeclared simpler-model tie-break. The validation errors tie, so the stated second $a = ry$ rule decides without consulting final data.

Changed-case reasoning: Select A using the predeclared simpler-model tie-break. The validation errors tie, so the stated second $a = ry$ rule decides without consulting final data.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Role Assignment: sort twenty actions into train/validation/test, then group related examples before splitting.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

A stronger claim makes a stronger project: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: Choose P with validation MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Train fits parameters and learned transforms; validation chooses settings; final test evaluates the selected system. Group related records first.

Independent question

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, validation and test?

Expected independent answer

A is training, B is validation and C is final test. Keep C separate from fitting and model selection.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Validation and final test data have interchangeable purposes during development.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Image counts are 54,18,18. Fit the mean on training images only, then apply the fixed transform to validation and test. Validation chooses settings; final test evaluates the selected system. Do not compute the mean from all 90 images.

Explanations for the additional workbook examples

W1 Choose a candidate

Apply the worked selection rule. Explain why B is not chosen.

Teacher explanation: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

W2 Assign data roles

Where should parameter fitting, setting selection and final reporting occur?

Teacher explanation: Fit on training, select on validation, evaluate the frozen choice on held-out test; fit preprocessing only on training.

W3 Record another attempt

A team creates D after reading validation mistakes. What changes in the record?

Teacher explanation: Add D and its motivation to development history. Validation influenced D and is not untouched test evidence.

W4 Apply independently

Training errors P 2 Q 0 R 1; validation errors P 4 Q 6 R 4. Rule: smallest validation MAE, then simpler; P is simpler than R. Choose and state the test action.

Teacher explanation: Choose P with validation MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

W5 Limit a claim

Why is the winning validation score not a guaranteed future score?

Teacher explanation: Selection used this sample, repeated attempts can adapt to it, and future conditions and sampling may differ.

Assess independent transfer

Selected case: Training errors P 2 Q 0 R 1; validation errors P 4 Q 6 R 4. Rule: smallest validation MAE, then simpler; P is simpler than R. Choose and state the test action.

Selected case answer

Teacher explanation: Choose P with validation MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Publish the grouped split strategy and counts for the Plant Classifier, including why plant identity stays within one split.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 25: Metrics Match Consequences. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use validation and hyperparameter in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Choose P with validation MAE 4; Q loses with 6. Keep the test unopened until settings and reporting rules are frozen. Fresh retry: validation L 2 M 2 N 3, simplicity M before L. Choose M under the declared tie policy; do not use test scores to break the tie. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Final testing and honest claims. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/25, ai-4/27, ai-4/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Select a model using a declared validation rule

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Where should parameter fitting, setting selection and final reporting occur?

Reasoned practice answer: Teacher explanation: Fit on training, select on validation, evaluate the frozen choice on held-out test; fit preprocessing only on training.

Check every learner before discussion: Use the worked case for Validation and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A and B both have validation MAE 2 cm; A is simpler under a predeclared tie rule. Require a prediction and a reason before comparing. Expected result: Select A using the predeclared simpler-model tie-break. Reason: The validation errors tie, so the stated second a = ry rule decides without consulting final data.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.